

## RESEARCH ARTICLE SUMMARY

## PLANT GENETICS

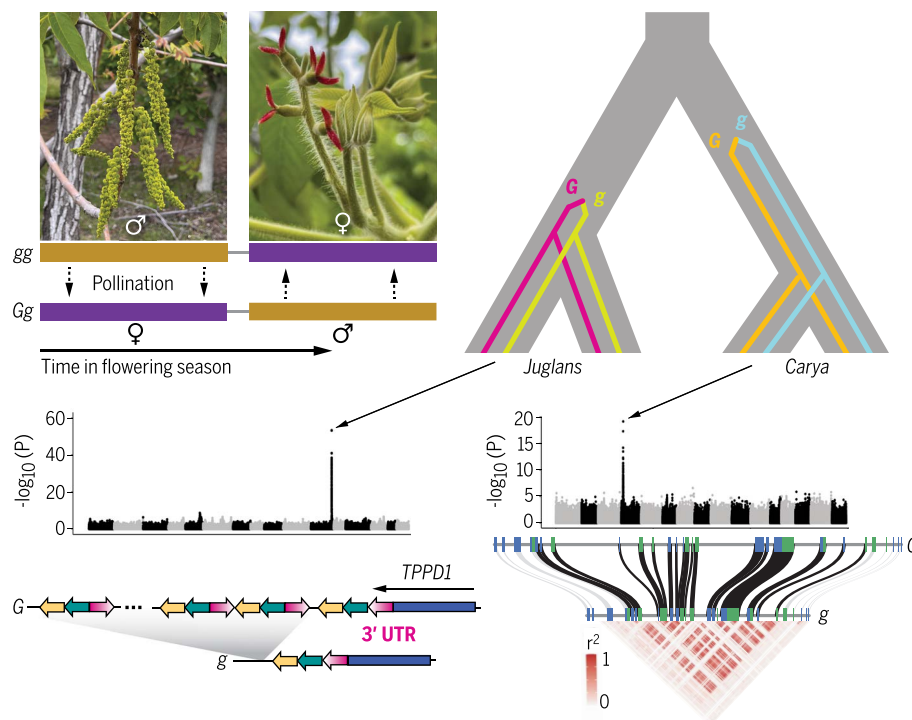
## Ancient structural variants control sex-specific flowering time morphs in walnuts and hickories

Jeffrey S. Groh\*, Diane C. Vik, Matthew Davis, J. Grey Monroe, Kristian A. Stevens, Patrick J. Brown, Charles H. Langley, Graham Coop\*

**INTRODUCTION:** Despite the capacity of many hermaphroditic plants to self-fertilize, the genetic costs of inbreeding have repeatedly driven the evolution of outcrossing strategies such as dioecy. Dioecy and other discrete mating type systems are maintained by rare mating type advantage and are often controlled by sets of tightly linked epistatic alleles (i.e., supergenes). We investigate the genetic underpinnings and evolution of a dimorphic mating system that exploits the temporal dimension - heterodichogamy. In species of hermaphroditic walnuts (*Juglans*) and hickories (e.g., pecans, *Carya*), two morphs alternate staminate and pistillate (i.e. male and female) flowering phases across the season,

promoting disassortative mating that maintains a stable equilibrium. The genetic underpinnings and evolution of this mating system have remained elusive.

**RATIONALE:** Heterodichogamy is known to be inherited through a single locus system in both walnuts and hickories (Juglandaceae), with a dominant allele for female-first flowering (*G*, dominant; *g*, recessive). These patterns suggested that heterodichogamy could have evolved in their common ancestor. We investigated the molecular basis of the inheritance of heterodichogamy in these genera, its evolutionary origins and maintenance over deep time scales, and its genetic parallels to dioecy.



**Multiple genetic mechanisms control a balanced polymorphism for flowering order in walnuts and hickories.** In *Juglans* (top left) and *Carya*, two morphs show complementary temporal separation between male and female flowering (heterodichogamy). Mating types are controlled by two nonhomologous single-locus mechanisms that arose in the common ancestor of each genus, respectively. (Bottom left) Simplified schematic of a putative functional regulatory element at the *Juglans* locus. (Bottom right) Strong genotypic correlations across the *Carya* locus indicate a lack of recombination between two colinear haplotypes with similar gene content.

**RESULTS:** We identified a narrow (~20-kb) genomic region controlling flowering types in Northern California black walnut (*Juglans hindsii*), and further found this region to be associated with flowering type across the entire walnut genus, spanning ~40 million years of evolutionary history. This region harbors a clear candidate gene (*TPPD1*) linked to trehalose-6-phosphate metabolism, a known flowering time pathway. Patterns of shared polymorphism across *Juglans* localize to the 3' untranslated region (UTR) of *TPPD1* and an adjacent tandem array of 8 to 12 inverted 3' UTR paralogs found only in the dominant haplotype. Presence of the dominant haplotype is associated with tissue-specific increased *cis* expression of *TPPD1* in early development of male flowers, predicting their delayed maturity. We identified a set of small RNAs transcribed from this tandem array at the same developmental stage and suggest that these may play a role in regulating *TPPD1*.

In pecan (*Carya illinoensis*), we identified a separate 200 to 445-kb region controlling heterodichogamy. This region shows strong linkage disequilibrium spanning ~20 protein-coding genes, several of which are linked to flower development. These haplotypes segregate throughout the *Carya* genus, and we estimate they diverged >50 million years ago. A lack of recombination in the near-permanently heterozygous dominant haplotype is accompanied by reduced genetic diversity and an accumulation of repetitive content, mirroring Y chromosome-like evolution. Neither the *Juglans* nor *Carya* locus segregates in other heterodichogamous genera within Juglandaceae, suggesting the existence of additional mechanisms.

**CONCLUSION:** Heterodichogamy can evolve through simple inheritance mechanisms that co-opt existing regulatory pathways and genetic architecture. These two distinct mechanisms in *Juglans* and *Carya* are deeply conserved but reflect either convergent evolution of the mating system or turnover of its control from an ancestral genetic system. Although these mechanisms are parallel to the genetics of dioecy, conservation of hermaphroditic function in heterodichogamy may limit the evolutionary advantage of recombination suppression thought to underlie sex chromosome heteromorphism. In summary, genetic mechanisms for heterodichogamy showcase both dynamic evolution and considerable stability. ■

The list of author affiliations is available in the full article online.

\*Corresponding author. Email: jgroh@ucdavis.edu (J.S.G.); gmcoop@ucdavis.edu (G.S.C.)

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## Ancient structural variants control sex-specific flowering time morphs in walnuts and hickories

Jeffrey S. Groh<sup>1,2\*</sup>, Diane C. Vik<sup>1</sup>, Matthew Davis<sup>3</sup>, J. Grey Monroe<sup>3</sup>, Kristian A. Stevens<sup>1,4</sup>, Patrick J. Brown<sup>3</sup>, Charles H. Langley<sup>1,2</sup>, Graham Coop<sup>1,2\*</sup>

Balanced mating type polymorphisms offer a distinct window into the forces shaping sexual reproduction strategies. Multiple hermaphroditic genera in Juglandaceae, including walnuts (*Juglans*) and hickories (*Carya*), show a 1:1 genetic dimorphism for male versus female flowering order (heterodichogamy). We map two distinct Mendelian inheritance mechanisms to ancient (>37 million years old) genus-wide structural DNA polymorphisms. The dominant haplotype for female-first flowering in *Juglans* contains tandem repeats of the 3' untranslated region of a gene putatively involved in trehalose-6-phosphate metabolism and is associated with increased *cis* gene expression in developing male flowers, possibly mediated by small RNAs. The *Carya* locus contains ~20 syntenic genes and shows molecular signatures of sex chromosome-like evolution. Inheritance mechanisms for heterodichogamy are deeply conserved, yet may occasionally turn over, as in sex determination.

Flowering plants have evolved a considerable diversity of reproductive strategies that offer insights into the adaptive value of sex and outcrossing (1). Most flowering plants are hermaphroditic and consequently, sexual reproduction comes with potential costs related to self-pollination (inbreeding) and conflict between male and female gamete success (sexual interference) (2, 3). A key insight, therefore, is that the precise developmental coordination of floral organs, i.e., stamens (male) and pistils (female), often reflects adaptation to minimize these costs (4). Mating systems with discrete morphs such as dioecy and heterostyly further illustrate that such adaptations, together with rare mating type advantage, can maintain stable floral polymorphisms over vast time scales (5–9).

Mating systems also exploit time. Dichogamy, the temporal separation of male and female sexual function within a hermaphrodite, is common among angiosperms and broadly viewed as a mechanism to limit sexual interference and inbreeding (10, 11). However, increasing temporal separation of male and female flowering within individuals is offset by the cost of increased temporal mismatch between reciprocal gamete types of potential mates (12). Heterodichogamy, first described in the cultivated Persian walnut (13), presents a potential resolution of this conflict in the form of stable dimorphism (Fig. 1B). In Darwin's words, "certain individuals mature their pollen

before the female flowers on the same plant are ready for fertilization and are called proterandrous [male-first]; whilst conversely other individuals, called proterogynous [female-first], have their stigmas mature before their pollen is ready" (1). This system generates strong disassortative mating between morphs (14), thus classical sex ratio theory predicts that rare mating type advantage drives populations toward a 1:1 equilibrium ratio of the two morphs (15).

Heterodichogamy has convergently evolved in several groups of angiosperms (16, 17), highlighting a clear adaptive benefit, yet it is puzzling why it is not more common (12), suggesting there could be constraints on its evolution. Since male and female flowering time must show different evolutionary responses during the establishment of two reciprocally synchronous morphs, one possibility is that multiple linked mutations might be required. Thus, identifying and characterizing the genetic loci controlling heterodichogamy across taxa can offer insight into its evolutionary origins and maintenance, and more generally, the nature of genetic variation for temporal sex allocation. Uncovering the genetic basis of heterodichogamy can further provide a useful comparative framework for investigating the processes underlying the evolution of heteromorphic sex chromosomes and other "supergenes" that control analogous dimorphic mating systems (18–20).

Heterodichogamy is well-known in several genera within Juglandaceae, the plant family that includes both walnuts (*Juglans*, Fig. 1, A and C) and pecans and other hickories (*Carya*) (13, 15, 21–23). It has been suggested as the ancestral state of subfamily Juglandoideae (24), which in addition to *Juglans* and *Carya* includes heterodichogamous genera *Platycarya*

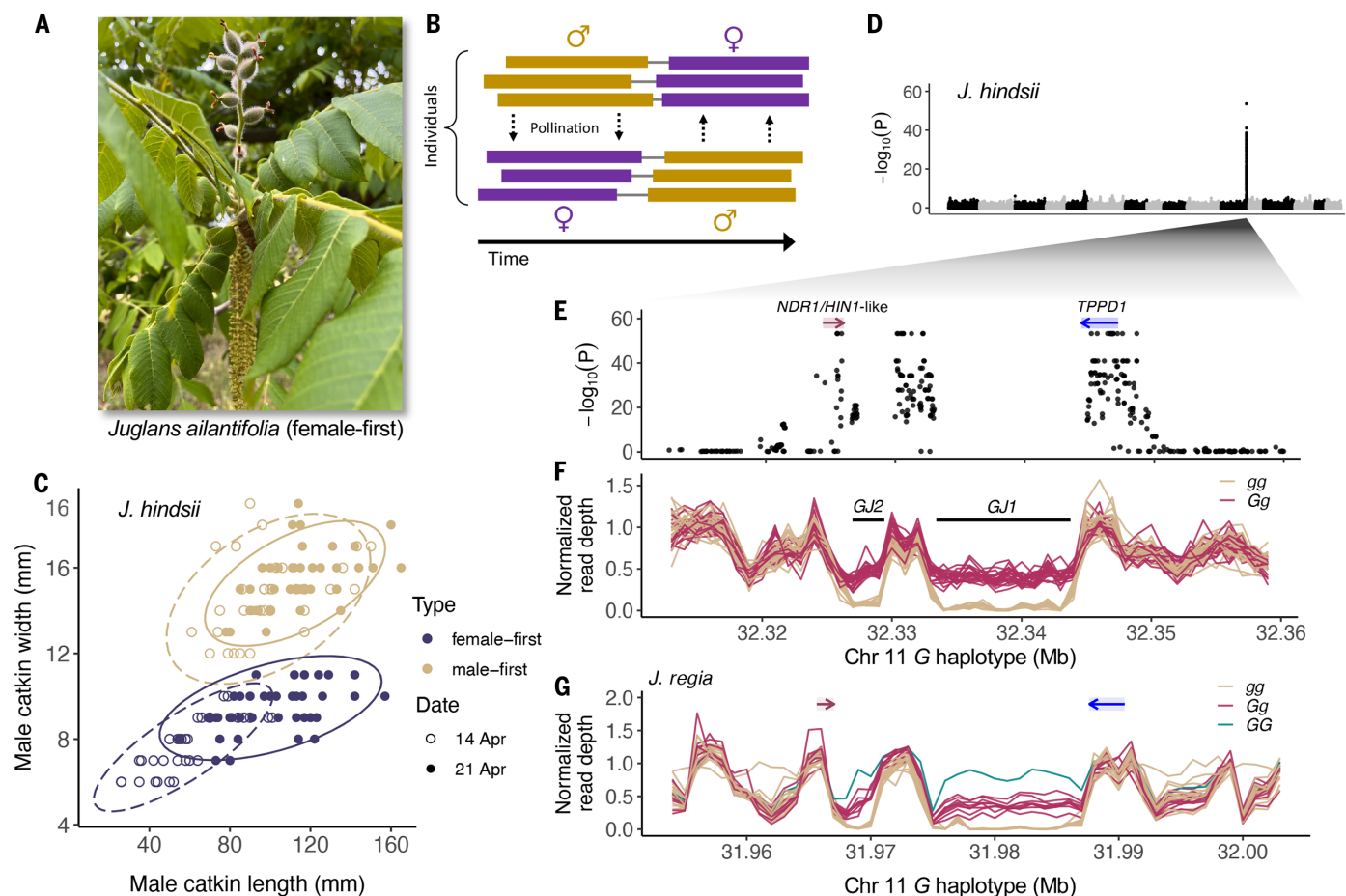
and *Cyclocarya* (24, 25). Fossil evidence indicates that the ancestor of these genera evolved unisexual, wind-pollinated flowers near the Cretaceous–Paleogene boundary [66 million years ago (Mya)] (26, 27), and subsequently radiated during the Paleocene (66 to 56 Mya) (28). An outgroup within Juglandaceae shows a form of monomorphic dichogamy (29), suggesting that heterodichogamy in Juglandoideae could have derived from monomorphic dichogamy. In both *Juglans* and *Carya*, heterodichogamy is controlled by a Mendelian locus with two alleles [*G* and *g*, following (15, 16)], in which the dominant allele *G* confers female-first flowering (15, 23). Using a comparative approach in these two related genera, we map and characterize the loci controlling heterodichogamy to elucidate its genetic basis, its origins and maintenance, and genetic parallels to other dimorphic mating systems.

Ancient regulatory divergence controls heterodichogamy across *Juglans*

In a natural population of Northern California black walnut (*J. hindsii*), male-first and female-first morphs were found to occur in roughly equal proportions (frequency of female-first morphs is 0.47, 95% confidence interval: 0.42 to 0.64,  $n = 81$ ) and show quantitative differences in their floral development (Fig. 1C). To map the genetic basis of the trait and characterize *G*-locus alleles, we generated haplotype-resolved chromosome-level assemblies from a female-first *J. hindsii* individual ("Rawlins", *Gg*) using PacBio HiFi sequencing (materials and methods). Likewise, to complement existing long-read assemblies of *J. regia g* haplotypes (30, 31), we generated haplotype assemblies from *J. regia* "Sharkey" a known homozygote for the dominant allele (*GG*) (15). We performed a genome-wide association study (GWAS) for flowering type using whole-genome resequencing data from 44 phenotyped *J. hindsii* (mean depth 46x). The GWAS revealed a single, strong association peak on chromosome 11, consistent with the known Mendelian inheritance of the trait in *Juglans* (Fig. 1D). This region is syntenic with the top GWAS hit location for dichogamy type in *J. regia* (fig. S1) (32, 33), as well as a broad region associated with dichogamy type in *J. nigra* (34). The association peak extends roughly 20 kb (hereafter, the *G*-locus) across a *NDRI/HINI*-like gene (hereafter *NDRI*-like) and a probable trehalose-6-phosphate phosphatase gene (hereafter *TPPD1*) (Fig. 1E). Several observations support the latter as a strong functional candidate. Trehalose-6-phosphate phosphatase (TPP) enzymes catalyze the dephosphorylation of trehalose-6-phosphate to trehalose as part of a signaling pathway essential for establishing regular flowering time in *Arabidopsis* (35) and inflorescence architecture in maize and grape (36, 37). *TPP* genes were recently identified as differentially expressed

<sup>1</sup>Department of Evolution and Ecology, University of California, Davis, CA, USA. <sup>2</sup>Center for Population Biology, University of California, Davis, CA, USA. <sup>3</sup>Department of Plant Sciences, University of California, Davis, CA, USA. <sup>4</sup>Department of Computer Science, University of California, Davis, CA, USA.

\*Corresponding author. Email: jgroh@ucdavis.edu (J.S.G.); gmcoop@ucdavis.edu (G.C.)



**Fig 1. A single locus controls heterodichogamy in *Juglans*.** (A) A female-first individual of *Juglans ailantifolia* with developing fruits and catkins (clusters of male flowers) shedding pollen. Photo by J.S.G. (B) Diagram of mating patterns under heterodichogamy. Male-first (top three) and female-first (bottom three) morphs show synchronized male and female flowering phases in reverse order of one another and which mate disassortatively across the flowering season. (C) Male catkins of male-first (tan) morphs of *J. hindsii* reach maturity and release pollen before those of female-first morphs (dark blue). Open circles and dotted ellipses show catkin sizes in one week during the first phase of the flowering season; filled circles and solid ellipses show sizes one week later.

(D) GWAS for dichogamy type in 44 individuals of *J. hindsii* from a natural population reveals a single association peak on chromosome 11. (E) The GWAS peak extends across the 3' region of *TPPD1*, a probable trehalose-6-phosphate phosphatase gene (blue rectangle with arrow indicating direction of transcription). Red rectangle shows the position of a *NDR1/HIN1*-like gene. (F) Normalized average read depth in 1-kb windows for 44 *J. hindsii* from a natural population against the same assembly. Black bars show positions of identified indels. (G) Normalized average read depth in 1-kb windows for 26 *J. regia* individuals at the G-locus. Colored rectangles indicate positions of genes orthologous to those in (E).

across the transition to flowering in both *J. sigillata* and *J. mandshurica* (38, 39). In *J. regia*, *TPPD1* is expressed in a broad range of vegetative and floral tissues (fig. S2) and we detected full-length transcripts from multiple tissues related to flowering in *J. regia* homozygotes for both G-locus haplotypes (table S1).

Consistent with a shared mechanism of inheritance across *Juglans*, we see similar patterns of copy number variation distinguishing two haplotypes at the GWAS peak in both *J. hindsii* and *J. regia* (Fig. 1, F and G, and fig. S1). In *J. hindsii*, all female-first individuals ( $n = 22$ ) are hemizygous across several >1-kb indels distinguishing *G* and *g* haplotypes whereas all male-first individuals ( $n = 22$ ) are homozygous for an indel allele specific to the *g* haplotype. The complete absence of any homozygotes for

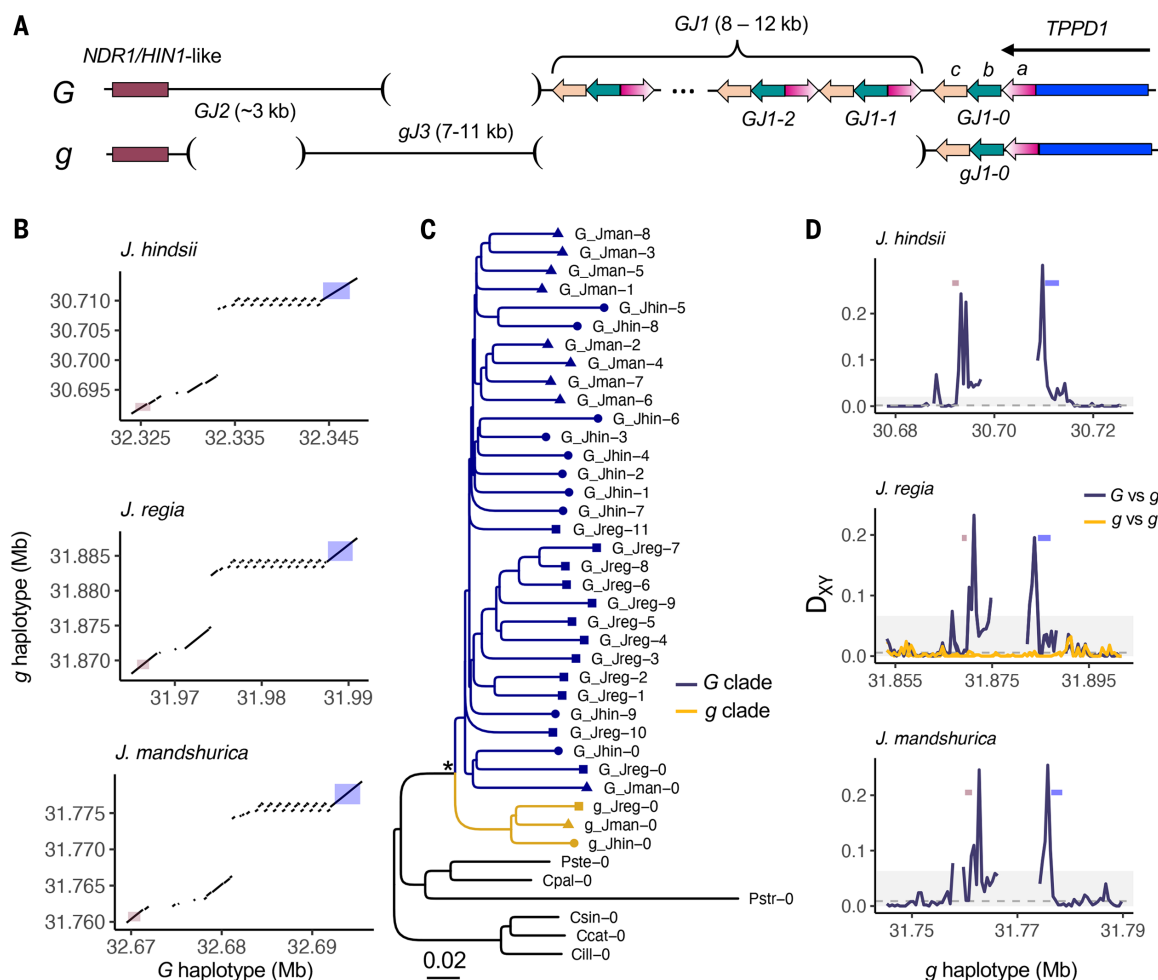
the *G* haplotype in our *J. hindsii* sample is consistent with disassortative mating at the G-locus in a natural population ( $P = 0.027$  for null hypothesis of Hardy-Weinberg proportions,  $\chi^2 = 4.89$ , d.f. = 1). These patterns are replicated in *J. regia* and additionally confirm the genotype of a known *GG* homozygote (“Sharkey”) (15), though *GG* homozygotes are rare (<2%,  $P = 0.021$  under null hypothesis of Hardy-Weinberg proportions,  $\chi^2 = 5.30$ , d.f. = 1) in a broad geographic sampling of this species. We generated whole-genome resequencing data from individuals of each dichogamy morph in five additional *Juglans* species (table S2) and examined public sequence data from known morphs in two additional species (40). Parallel patterns of read depth support the presence of a shared structural variant segregating in perfect asso-

ciation with dichogamy type across all nine species examined (fig. S3). In addition, we identified both G-locus haplotypes represented in existing long-read assemblies from several *Juglans* species. The presence of shared haplotype structure across the genus suggests that these haplotypes could have diverged in the common ancestor.

Nucleotide divergence between G-locus haplotypes is exceptionally high compared with the genomic background in several species, supporting an ancient divergence between these haplotypes (Fig. 2D). The level of divergence is roughly comparable to the divergence observed between *Juglans* and *Carya* in the same region (fig. S4). Applying a molecular clock approach, we estimated the age of the G-locus haplotypes to be 37.3 to 40.2 Mya (materials

**Fig. 2. Molecular structure and divergence at the *Juglans* G-locus.**

(A) Schematic of the G-locus structural variant in *Juglans*. Presence or absence of indel alleles is indicated by continuous lines versus open parentheses. Maroon and blue bars show bordering genes. The black arrow across *TPPD1* indicates direction of transcription. Colored arrows represent subunits of the repeat motif within the *GJ1* insertion and paralogous sequence outside of the indel. Each repeat motif (numbered *GJ1-1*, *GJ1-2*, ..., *GJ1-n*) is comprised of subunits a, b, and c. Subunit a, which is homologous with the 3' UTR of *TPPD1*, is inverted within the insertion. (B) Dotplots showing pairwise alignments of alternate G-locus haplotypes for three species representing the major clades within *Juglans*. Maroon and blue rectangles indicate the locations of bordering genes as in (A). (C) Phylogeny of *GJ1* repeats. Sample codes from left to right indicate: haplotype (for *Juglans* only), abbreviated binomial, and repeat number [as in (A)] (see fig. S8 for larger phylogeny and table S3 for sample codes). Sequences from *g* haplotypes (gold) and *G* haplotypes (blue) form two sister clades (asterisk indicates 96% bootstrap support) whose most recent common ancestor predates the radiation of *Juglans*. Node shapes represent species. Branch lengths reflect substitutions per site. (D) Average nucleotide divergence in 500-bp windows between alternate G-locus haplotypes for three within-species comparisons. Dotted gray lines indicate the chromosome-wide average and shaded gray regions indicate 99% quantiles. Maroon and blue bars indicate positions of genes as in (A).



and methods). As both fossil and molecular evidence place the most recent common ancestor of *Juglans* in the Eocene (33.9 to 55.8 Mya) (28, 41, 42), this estimate is consistent with G-locus haplotypes diverging in the common ancestor of extant *Juglans*.

To identify candidate functional variants under long-term balancing selection, we focused on haplotype sequence differences conserved across the entire *Juglans* phylogeny, beginning with the *TPPD1* gene sequence. In a comparison of both haplotypes from three species spanning the earliest divergence events within *Juglans*, we find several conserved single nucleotide polymorphisms (SNPs) within the 3' untranslated region (UTR) of *TPPD1* (fig. S5A) that consistently group by haplotype rather than by species. Despite high linkage disequilibrium (LD) across the coding sequence of *TPPD1* within species (further than expected given the background rate of LD decay) (fig. S6), polymorphism within the coding region of *TPPD1* groups by

species rather than haplotype, indicating recombination over deep time scales and a lack of shared functional coding divergence (fig. S5A). However, numerous coding polymorphisms are shared among more closely related species (North American black walnuts) (fig. S5B). This observation suggests the possibility of recruitment and turnover of functional divergence within the *TPPD1* coding sequence. Similarly, a small number of fixed haplotype differences restricted to the 3' UTR of *NDR1*-like are conserved across more closely related species, but none across the whole genus (fig. S5C). Notably, *NDR1*-like is expressed in *J. regia* flowers (fig. S2B), suggesting that it too might play a role in morph specialization. Nonetheless, the lack of genus-wide conserved coding polymorphism suggests that the primary functional differences may be regulatory.

We then investigated the indels distinguishing *G* and *g* haplotypes (*GJ1* and *GJ2* indel alleles present in the *G* haplotype; *gJ3* indel

allele present in the *g* haplotype, Figs. 1F, 2A, and S1). The indel alleles *GJ2* and *gJ3* are not obvious functional candidates. For example, *gJ3* is derived from an insertion of a CACTA-like DNA transposon in *g* haplotypes of *Juglans*, which lacks evidence of strong sequence conservation across species (fig. S7). By contrast, *GJ1* is more suggestive of possible regulatory function. Using pairwise alignments, we found that *GJ1* contains a series of ~1-kb tandem repeats that are paralogous with sequences in the 3' region of *TPPD1* (Fig. 2, A and B). We find no evidence for the inclusion of the repeat structure in full length *TPPD1* transcripts from a *GG J. regia* individual. Each repeat, ordered *GJ1-1* up to *GJ1-n* moving downstream of *TPPD1*, consists of three subunits. One subunit (*GJ1a*) is an inverted ~250- to 300-bp (base pair) motif homologous with the 3' UTR of *TPPD1*. The other two subunits (*GJ1b* and *GJ1c*) are noninverted ~300-bp motifs homologous with regions within 1 kb downstream of the 3' UTR.

These tandem duplicates are present in *G* haplotypes across multiple species with long read assemblies, varying in number from 8 to 12 copies (Fig. 2B). A maximum likelihood phylogeny of concatenated aligned sequence from *GJI* repeats and their homologous sequence immediately downstream of *TPPD1* (*GJI-0/gJI-0*) shows that all *G* haplotype sequences form a sister clade to all *g* haplotype sequences (Fig. 2C, fig. S8), further evidence of haplotype divergence in the common ancestor of *Juglans*. Among more closely related species (North American black walnuts), we found evidence of conservation of specific repeats, with repeats in different species at the same relative positions in the array clustering together in the phylogeny (fig. S8). On the other hand, over deeper timescales, evolution of *GJI* repeats is characterized by lineage-specific turnover as a result of expansion and contraction of the repeat array and possible gene conversion. For example, the majority of repeats within *J. regia* are most closely related to repeats at other positions within the same species than to repeats in more distant relatives in the sample.

In Juglandaceae, floral initiation begins in the growth season prior to flowering. In *J. regia*, male-first and female-first morphs differ in the extent of floral development completed prior to dormancy (43, 44). Correspondingly, in *J. regia* we found that male catkin buds (clusters of developing male flowers, fig. S9A) of male-first morphs in this species are larger than those of female-first morphs by midsummer in the season prior to bloom (fig. S9B). Given the conserved non-coding *G* haplotype insertion, and the lack of *Juglans*-wide trans-specific coding polymorphism in *TPPD1*, we hypothesized that the developmental differential between dichogamy types may be due to differential regulation of *TPPD1* during early floral development. We thus performed RNA sequencing (RNA-seq) from male catkin bud tissue prior to dormancy for both morphs in both *J. hindsii* and *J. regia*. In both species, female-first morphs show higher expression of *TPPD1* compared

with male-first morphs ( $P = 8.71 \times 10^{-8}$ ,  $\chi^2 = 5.35$ , d.f. = 1, Fig. 3), and in *J. hindsii* *TPPD1* is among the top three most significant differentially expressed genes. Heterozygotes in both species show a strong allelic expression bias toward the *G* copy of *TPPD1*, and in *J. regia* the highest expression was seen in the *GG* homozygote “Sharkey,” consistent with conserved *cis*-acting regulatory divergence between *G*-locus alleles. We found extremely low expression of *NDR1*-like in only two *J. hindsii* samples (zero in all others), although we note this gene is expressed in later developmental stages of male catkins and in female flowers of *J. regia* (fig. S2B). We found the same pattern of allele-specific expression of *TPPD1* replicated in two *J. mandshurica* datasets (figs. S10 and S11) (39, 45), an independent *J. regia* dataset (fig. S12) (46), and in *J. nigra* (fig. S13) (47). Evidence for morph-specific regulation of *TPPD1* in female flowers across these datasets is mixed (figs. S10 to S13), perhaps due to a lack of sampling of relevant developmental stages. Notably, *TPPD1* alleles are not differentially expressed in vegetative tissues of heterozygotes, indicating that regulatory effects of *G*-locus variation on *TPPD1* expression are tissue-specific (fig. S13). In sum, increased expression of the *G* allele of *TPPD1* in early development of male flowers of female-first morphs is shared across the *Juglans* clade.

To explore possible mechanisms by which variation at the *G*-locus might regulate *TPPD1* expression, we generated small RNA (sRNA) sequence libraries from male catkin buds of both morphs in *J. hindsii* and *J. regia*. In both female-first *J. hindsii* and two out of three female-first *J. regia* samples, we found numerous (predominantly 24-nt) sRNAs that map to the *GJI* indel (figs. S14 and S15). We also examined a publicly available sRNA dataset (48) from the same tissue from both morphs of *J. mandshurica* individual and observed the same pattern (fig. S16). We identified one set of sRNAs whose sequences match the same region of *TPPD1* 3' UTR-homologous repeats across species (figs.

S17 and S18). Other suggestive candidates are species-specific (figs. S17, S19, and S20). Further investigation of the functional role of these sRNAs is clearly warranted, given an emerging view of the general importance of sRNAs in regulating plant sexual systems (49, 50). The transcription of sRNAs with possible regulatory function from *GJI* repeats, which vary in number across and within species, suggests the possibility that this molecular structure could facilitate quantitative, morph-specific evolutionary shifts in flowering time.

Taken together, multiple independent lines of evidence support that the control of dichogamy type in *Juglans* is governed by *TPPD1* regulatory variation that predates the radiation of the genus: (i) the genus-wide balanced polymorphism of the array of ( $\geq 8$ ) 3' UTR homologous repeats, (ii) the production of sRNAs by this *G* haplotype repeat array matching sites within and downstream of the *TPPD1* 3' UTR, (iii) and differences in *TPPD1* expression between dichogamy morphs due to higher relative expression of the *G* haplotype copy of *TPPD1* in developing male flowers. (iv) Finally, the presumed substrate of *TPPD1*, T6P, has been found in model systems to play a critical role in regulating the transition to flowering. In *Arabidopsis* for instance, overexpression of a heterologous *TPP* delays flowering (51), as does knockdown of a T6P synthase gene (35). These facts are consistent with the idea that higher expression of *TPPD1* in developing male flowers of female-first individuals could be responsible for delayed male flowering. Although the mechanistic basis of the expression difference is not clear, we note that small RNAs have been reported to upregulate gene expression in some cases (52, 53).

Finally, we note that while the divergence of *G*-locus haplotypes predates the *Juglans* radiation, our molecular dating and phylogeny inference support a haplotype divergence more recent than the divergence of *Juglans* from its closest cousins, *Pterocarya* and *Cyclocarya*, and the relationships of non-*Juglans* sequences resemble previous estimates of the species tree for these taxa (Fig. 2C). Furthermore, we found no evidence of the *G*-locus structural variant in resequencing data from multiple individuals from each of *Cyclocarya*, *Pterocarya*, and *Platycarya* (fig. S21).

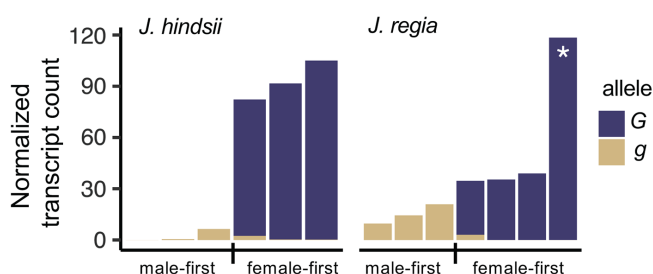
### A supergene controls heterodichogamy across *Carya*

Although heterodichogamy has been suggested as the ancestral state of subfamily Juglandoideae (24), our results indicate that the *Juglans* *G*-locus is specific to walnuts. To identify the basis of the trait in pecan (*Carya illinoensis*), we generated whole-genome resequencing data from 18 pecan varieties of known dichogamy type (table S2) and combined these with existing data for 12 additional varieties (54) to perform

**Fig. 3. Differential regulation of *TPPD1* in developing male flowers of *Juglans*.**

*TPPD1* expression in developing male flowers is higher in female-first morphs relative to male-first morphs. Heterozygotes for the *G*-locus show higher relative expression of the *G* allele copy of *TPPD1*. The highest

expression is seen in a homozygote for the *G* allele in cultivated Persian walnut (*J. regia* “Sharkey”), marked with an asterisk. The other three female-first *J. regia* samples were confirmed as heterozygotes (*Gg*), and male-first samples were confirmed as *gg* homozygotes. Two female-first *J. hindsii* samples were confirmed as *Gg* heterozygotes (the other is a presumed heterozygote), and two male-first samples were confirmed as *gg* homozygotes.



**Fig. 4. The G-locus in**

**Carya.** (A) GWAS for dichogamy type in 30 pecan varieties identifies a single peak of strong association on chromosome 4.

(B) Zoomed-in view of the GWAS peak. Rectangles (bottom of plot) indicate positions of predicted genes. Red rectangles indicate those that fall within the region of strong association.

(C) Normalized average read depth in 1-kb windows across the location of the GWAS hit reveals numerous copy number variants co-segregating in perfect association with dichogamy type.

(D) Nucleotide divergence between G-locus haplotypes within coding sequence is strongly elevated against the genome-wide background.

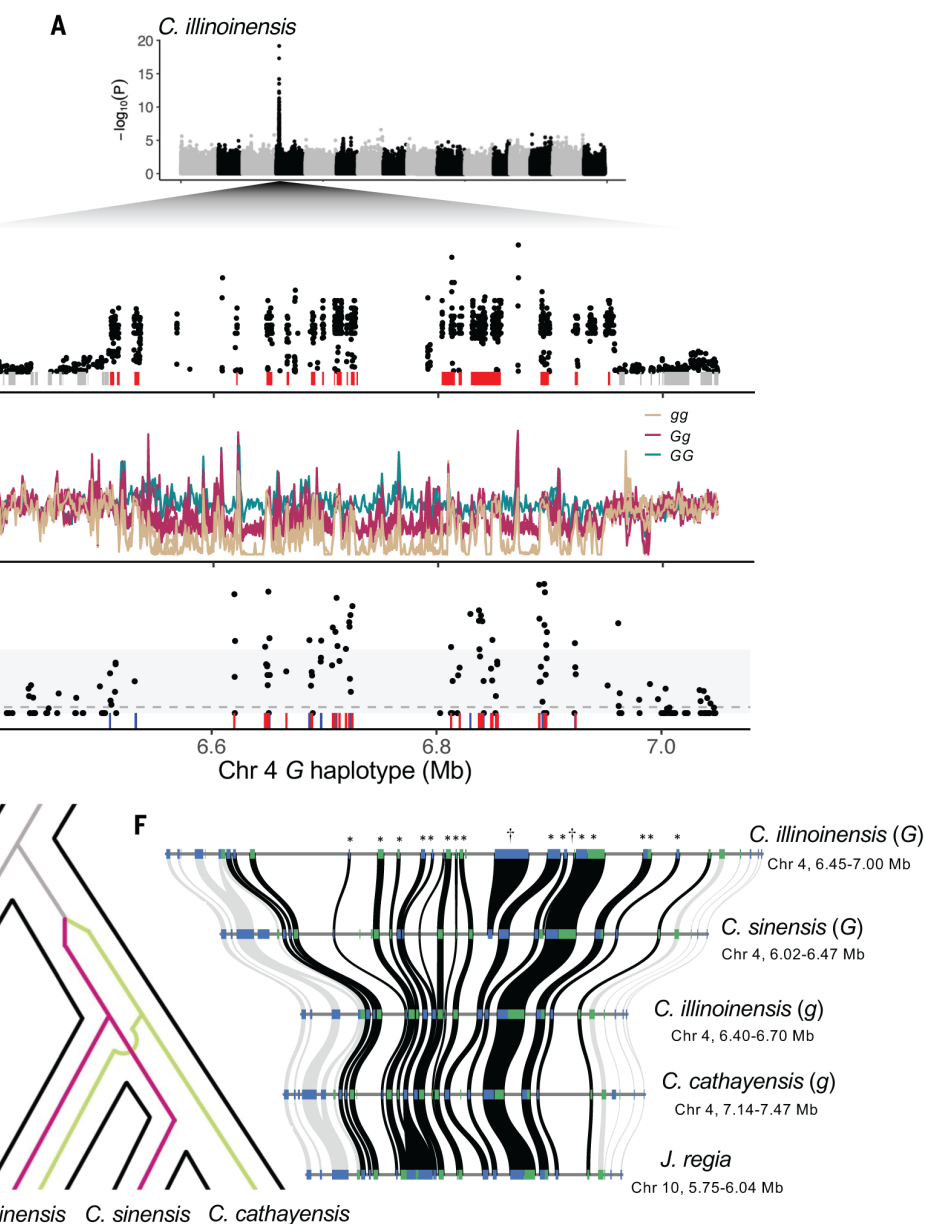
Dotted line shows the genome-wide average for coding sequence and the shaded interval shows 99% quantile. Colored tick marks (bottom) show locations where other *Carya* species are polymorphic for SNPs that are fixed between pecan G-locus haplotypes.

Blue, North American; red, East Asian. (E) The phylogeny of the G-locus is discordant with the species tree, reflecting trans-species polymorphism.

*Carya* G-locus haplotypes (G, pink; g, green) diverged after the split with *Juglans* and were present in the most recent common ancestor of *Carya*.

(F) Gene-level synteny for assemblies of both G-locus haplotypes in North American pecan, East Asian *Carya*, and *J. regia*. G haplotypes are longer than g haplotypes in both major *Carya* clades as a result of greater transposable element content. Colored rectangles show positions of genes in each assembly, with color indicating strand direction (blue, forward; green, reverse). Black bands connect orthologous genes that fall within the GWAS peak in pecan; gray bands connect genes outside the region. Individual gene trees that support the trans-species polymorphism are indicated by asterisks along the top. Daggers indicate two predicted genes that appear to be shared by North American and East Asian G haplotypes.

a GWAS for dichogamy type. We identified a single strong association peak on chromosome 4 (Fig. 4A), consistent with previous quantitative trait locus mapping (55), with many alleles in an unusually large LD block compared with the genomic background (fig. S22). This pecan G-locus region is not homologous with the *Juglans* G-locus (it involves different genes), implying either convergent origins of hetero-dichogamy within Juglandaceae or a single origin followed by turnover in the underlying genetic mechanism.



We identified numerous copy number variants [spanning 445 kb in the primary haplotype assembly of “Lakota” (56), Gg diploid genotype] overlapping the GWAS peak that segregate in perfect association with dichogamy type (Fig. 4C) and which confirmed the genotype of two known GG homozygotes, “Mahan” and “Apache” (23, 55). Coverage patterns in this region indicate that the primary assembly of “Lakota” contains the G haplotype. Consistent with this, coverage patterns against the alternate assembly of “Lakota” and against the reference

assembly of “Pawnee” (male-first) identified both as containing g haplotypes (fig. S23).

In contrast to the *Juglans* G-locus, the *Carya* G-locus contains approximately 20 predicted protein-coding genes (Fig. 4B, red rectangles, and table S4), and gene-level synteny is highly conserved between the two haplotypes with no inversion detected (Fig. 4F). Segregating copy number variation is largely localized to intergenic regions, though we note several small indels within coding sequence (fig. S24). Several of these genes have homologs with

established functional roles in flowering (e.g., *FIL1*, *EMS1*, *SLK2*, *CEN/TFL1*; see table S4) and are promising candidates for regulating the development of alternate dichogamy types. We found 237 nonsynonymous fixed differences between the pecan haplotypes across these 20 genes, suggesting many potential functional candidates. However, applying the McDonald-Kreitman test to each gene, the ratio of non-synonymous to synonymous substitutions versus polymorphisms did not identify any signals of adaptive evolution. We note however that two of these genes with annotated functional roles in flowering, *EMS1* and *FIL1*, were previously identified as being in a small set of highly differentially expressed genes in male catkin buds between a male-first and female-first cultivar during early flower development (57). *Arabidopsis* mutants of *EMS1* fail to form pollen tetrads (58), and female-first pecan varieties have a delayed progression from pollen tetrads to mature pollen as compared with male-first varieties (22). *FIL1* homologs are expressed uniquely or predominantly in stamens (e.g., fig. S25) (59) and were found to be regulated by the B class MADS-box transcription factor *DEFICIENS* in *Antirrhinum majus* (59) (the ortholog of *Arabidopsis AP3*).

Nucleotide divergence between *Carya* G-locus haplotypes is strongly elevated against the genome-wide background, suggesting that they could also have an ancient origin (Fig. 4D). The *Carya* genus shows marked range disjunction between North American and East Asian subclades, which diverged from one another approximately 22 Mya (60) (Fig. 4E). Patterns of read depth of the North American pecan varieties against two published genome assemblies from East Asian *Carya* (61) suggested that syntenic regions in assemblies of *C. cathayensis* and *C. sinensis* represent the *g* and *G* haplotypes, respectively (fig. S26). This was further supported by elevated nucleotide divergence between the assemblies in this region (fig. S27) and shared nonsynonymous coding polymorphisms with pecan haplotypes (fig. S24). We next constructed a maximum likelihood phylogeny using concatenated coding sequence alignments from the 20 G-locus genes, and found that it conflicts with the species tree, with *C. cathayensis* clustering with the pecan *g* haplotype and *C. sinensis* clustering with the pecan *G* haplotype (both clades 100% bootstrap support) (Fig. 4E). Finally, we used whole-genome resequencing data from 16 individuals [15 from (62), 1 newly sequenced] representing 15 non-pecan *Carya* species across both East Asian and North American clades to examine heterozygosity at the G-locus across the genus. Out of 16 individuals in this sample, 8 were highly heterozygous (25 to 74% whereas the other 8 were 1 to 5%) for a set of 454 coding sequence SNPs ascertained in pecan. This pattern is consistent with these individuals being

heterozygous for G-locus haplotypes and with dichogamy types being maintained at close to 1:1 ratios in species across the entire genus (figs. S28 and S29). Thus, multiple lines of evidence indicate that the *Carya* G-locus haplotypes are at least as old as the divergence between North American and East Asian clades of *Carya*, a depth of ~22 Mya (60).

If variation at the *Carya* G-locus controlled dichogamy types in the ancestor of *Juglans* and *Carya*, we would expect that nucleotide divergence between the *Carya* G-locus haplotypes should match or exceed nucleotide divergence between *Carya* and the orthologous sequence in *Juglans*. Divergence between *Carya* G-locus haplotypes at fourfold degenerate sites across the core set of *Carya* G-locus genes that group by haplotype was less than that between either *Carya* haplotype and *J. regia* (0.053 versus 0.064), implying that the *Carya* G-locus haplotypes diverged more recently than the *Juglans-Carya* split. A maximum likelihood phylogeny across these genes rooted using valley oak as an outgroup (*Quercus lobata*) further supports *Juglans* as an outgroup to *Carya* haplotypes (98% bootstrap support, fig. S30). Using a molecular clock approach (materials and methods), we estimate the age of the *Carya* G-locus haplotypes to be 54.7 to 59.1 Mya.

The elevated scale of LD across the *Carya* G-locus is exceptional compared with the genome-wide average (fig. S22), mirroring the widespread pattern of suppressed recombination between sex chromosomes in many taxa. Theory and data suggest that recombination suppression between sex chromosomes can accrue progressively, perhaps driven by selection favoring co-transmission of sex-determining alleles and alleles with sexually antagonistic effects (18, 63). In the evolution of heterodichogamy, if separate mutations are needed to regulate female and male flowering time, fitness epistasis could select for reduced recombination between them. Although most of the gene trees across the G-locus support the North American-East Asian trans-species polymorphism, we note that the three genes at the left border (including the functional candidate *EMS1*) and the gene at the right border support the species tree for these taxa (Fig. 4E). Furthermore, we do not see clear evidence for trans-species polymorphisms within these genes in the East Asian clade (Fig. 4D and fig. S29). These observations, along with the lower divergence between pecan G-locus haplotypes in the three genes at the left border, suggest that the boundary of recombination suppression at the G-locus could have changed since the divergence of the North American and East Asian *Carya*, perhaps after the appearance of genetic variation in neighboring genes with antagonistic fitness effects between the two dichogamy morphs.

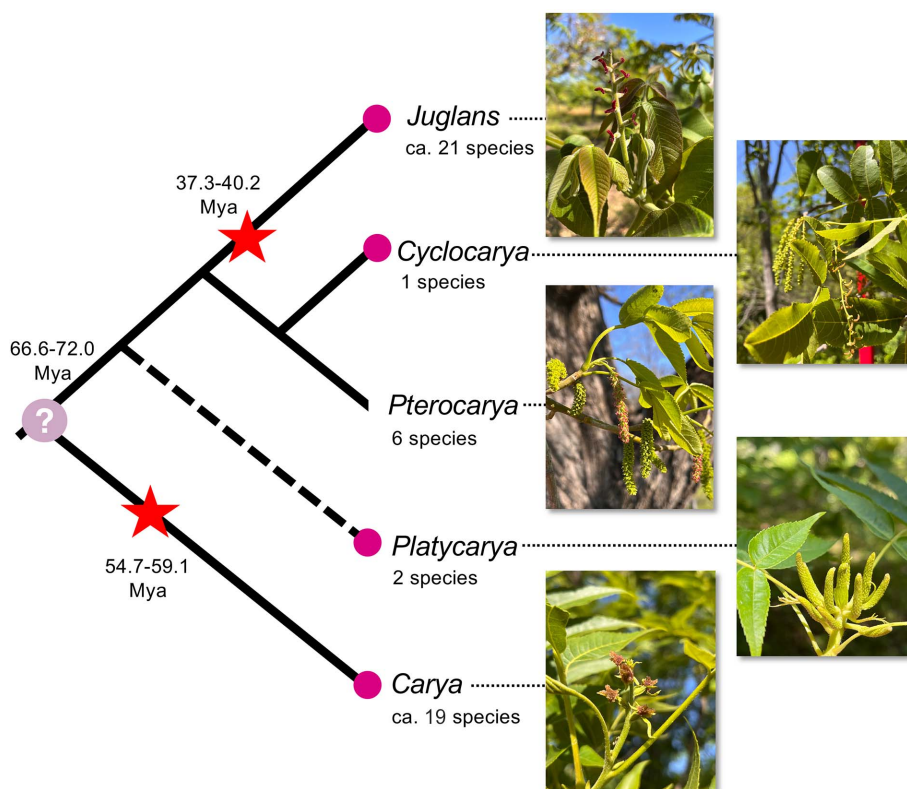
We also note a small number of predicted protein-coding genes at the G-locus that differ

in presence/absence among *Carya* assemblies (Fig. 4F and table S5). Notably, two of these appear to be derived and shared between North American and East Asian *G* haplotypes and so warrant further attention (table S5). However, broad synteny of 20 predicted protein-coding genes across *Carya* G-locus assemblies is conserved as far back as the divergence with oaks (~90 Mya, fig. S31) (64). This suggests that the establishment of G-locus haplotypes in *Carya* may have been facilitated by preexisting physical linkage of flowering genes that could serve as functional targets during adaptive evolution of two dichogamy morphs.

Non-recombining portions of sex chromosomes and other supergenes often accumulate more deleterious variation relative to recombining counterparts, in some cases leading to highly degenerate Y chromosomes (63, 65, 66). Notably, the pecan *G* haplotype (445 kb) is more than twice as long as the *g* haplotype (200 kb), largely due to a difference in transposable element (TE) content within the region of reduced recombination (*g*: 40%, *G*: 65%, genome average: 53%; fig. S32). Part of the transposable element expansion in the *G* haplotype may be ancient, given parallel coverage patterns (Fig. 4C and figs. S23 and S26) and assembly lengths for the East Asian G-locus haplotypes; the *C. sinensis* *G* assembly is ~350 kb whereas the *C. cathayensis* *g* assembly is ~230 kb. This proliferation of TEs within *G* haplotypes could reflect reduced efficacy of selection on the *G* haplotype as a result of its lower copy number (1 *G* for every 3 *g* haplotypes if disassortative mating is complete), and/or the enhanced effects of positive and negative linked selection due to its low recombination rate (64, 65). Consistent with both possibilities, we observe an approximately sixfold reduction in genetic diversity in coding regions for *G* haplotypes compared with *g* haplotypes ( $\pi_G = 0.0018$ ,  $\pi_g = 0.0111$ ;  $P < 0.013$ , permutation).

### Genetic systems for heterodichogamy in other genera

Our estimate of the age of the *Carya* G-locus haplotypes suggests that we do not expect to find these haplotypes segregating in heterodichogamous *Cyclocarya* and *Platycarya* (Fig. 5). Nonetheless, we considered the possibility that a subset of the *Carya* G-locus variation could be ancestral across heterodichogamous genera, or that there has been convergent use of the same region. Therefore, we examined heterozygosity at the region syntenic with the *Carya* G-locus in a sample of 11 *Pterocarya stenoptera*, 12 *Cyclocarya paliurus*, and 3 *Platycarya strobilaceae*, but we found no evidence of increased polymorphism in this region comparable to patterns seen in *Carya* (fig. S33). As our analyses also indicate that heterodichogamy in these genera is not controlled by the *Juglans* G-locus (Figs. 2C, S21,



**Fig. 5. Hypothesized evolutionary history of heterodichogamy within Juglandoideae.** Cladogram showing relationships among five genera within Juglandoideae, adapted from (41), with representative photos of floral structures of each. The phylogenetic placement of *Platycarya* remains debated and some studies place it as an outgroup within this clade. Photos (top to bottom): *Juglans ailantifolia*, *Cyclocarya paliurus*, *Pterocarya stenoptera*, *Platycarya strobilacea*, *Carya illinoensis*. Dark pink circles indicate genera with reported heterodichogamy in the literature; red stars indicate the inferred time of origin for the *Juglans* and *Carya* loci that control heterodichogamy. Heterodichogamy may have evolved convergently, or in the common ancestor (faded pink circle with question mark) followed by turnover in the genetic mechanism. Photos by J.S.G.

and S34), this suggests the possibility that additional genetic systems controlling heterodichogamy in other Juglandaceae genera remain undiscovered.

## Discussion

As balanced polymorphisms evolve, selection can favor reductions in recombination and changes to genome structure that heighten linkage between advantageous combinations of alleles, potentially driving the evolution of “supergenes” (18–20). Both the *Juglans* and *Carya* heterodichogamy loci show evidence of reduced recombination compared with the genomic background (figs. S6 and S22). Thus, the associated structural variants may have facilitated the evolution of this mating system in part through their effects on linkage between multiple functional mutations. Across *Juglans*, heterodichogamy is controlled by a narrow region partially overlapping a single flowering time candidate gene (*TPPD1*) that shows morph-specific regulation in male flowers. This region must also control the female flowering time difference between morphs.

Is a single pleiotropic change responsible? A pleiotropic change acting through expression of *TPPD1* in both male and female flowers would need to have opposing effects on their development, which seems unlikely. Alternatively, a change that delays male flowering through *TPPD1* could advance female flowering through a resource tradeoff. This also seems unlikely; in *Juglans*, male and female flowers are highly dimorphic structures that are borne in separate locations on the plant and respond differently to environmental flowering cues (67), suggesting that their development is not tightly coupled. Given these difficulties with the pleiotropy hypothesis, we consider it more likely that this region harbors additional *cis* regulatory changes affecting *TPPD1* (or possibly *NDR1*-like) that separately regulate female flowering time. Identifying the basis of the female flowering difference will be key to understanding the role of linkage in the evolution of heterodichogamy.

In *Carya*, where flowering types segregate for likely many functional variants across ~20 coding genes, a heterodichogamy supergene arose

in a genomic region with preexisting physical linkage among several flowering related genes that could be recruited into the control of dichogamy type. In the absence of an inversion, it is plausible that differential accumulation of TEs on the *G* haplotype may suppress recombination through effects on chromatin configuration (68); however, it is difficult to distinguish between the TEs as a cause versus consequence of recombination suppression (69). Indeed, elevated accumulation of TEs on the *Carya G* haplotype mirrors a common pattern seen in analogous systems of heterogametic sex determination, in which a lack of recombination over permanently heterozygous portions of the Y chromosome is accompanied by the accumulation of repetitive sequence presumed to be deleterious (66) (although recombination in rare *GG* homozygotes resulting from selfing or imperfect disassortative mating may retard this effect in heterodichogamy).

Although the *Juglans* and *Carya* heterodichogamy loci likely involve multiple functional changes, we also note that they are restricted to relatively small genomic regions, in contrast to some highly heteromorphic sex chromosomes of similar or younger age (70). This observation can be reconciled with a popular model for sex chromosome heteromorphism that suggests natural selection progressively drives extended crossover suppression between the sex-determining region and alleles with sexually antagonistic effects (18, 63, 71). In heterodichogamy, selection to maintain both male and female sexual functions in both dichogamy morphs likely acts as a constraint on the availability of physically linked epistatic variation that could trigger adaptive recombination suppression and subsequent evolution of haplotype heteromorphism. As heterodichogamy is thought to be one pathway to dioecy (16, 72, 73), comparative genomic studies in clades containing both mating systems could provide further tests of this hypothesis.

Despite considerable stability of two genetic systems for heterodichogamy within Juglandaceae, the genetic mechanisms underlying heterodichogamy appear to be more diverse than previously appreciated. Because neither the *Juglans* nor *Carya G*-locus is as old as their common ancestor (Fig. 5), our data cannot rule out the possibility that these genetic systems originated independently through the convergent evolution of heterodichogamy. However, given the prevalence of heterodichogamy across Juglandoideae, it also seems plausible that the mating system evolved once in the ancestor and has been maintained, but that genetic control of mating types has been subject to turnover. Similar dynamics are common in some clades with separate sexes (74, 75). In this scenario, both loci might represent turnover from an unknown ancestral genetic system

(Fig. 5). This suggests the prospect of investigating the frequency and evolutionary drivers of turnover by mapping additional heterodichogamy loci in a broader comparative framework.

## Materials and methods summary

We studied a natural population of *Juglans hindsii* in Davis CA, USA, over the course of the flowering season in spring of 2023. Individuals were scored as either female-first or male-first based on the relative developmental stages of the male and female flowers. We generated high-quality haplotype-resolved genome assemblies from a single female-first *J. hindsii* individual using PacBio HiFi reads assembled with *hifiasm* (76) and combined these with whole-genome resequencing from the natural population to perform fine-mapping of the G-locus (77) after standard bioinformatics processing steps (78, 79). We generated haplotype-level assemblies from a female-first *J. regia* individual using the same methods. We used several sequence alignment methods (78, 80, 82) to examine structural differences between *Juglans* haplotypes and identify regions of conserved polymorphism across species. Maximum likelihood phylogenies of homologous regions were inferred using IQ-TREE (83). We performed RNA-seq from male flower bud tissue and tested for differential expression of *TPPDI* between morphs using Salmon (84) and DESeq2 (85). We phenotyped and performed resequencing of pecan varieties to map the *Carya* G-locus against existing assemblies (56). We used sequence alignment methods and transposable element annotation with EDTA (86) to examine haplotype structure and sequence content, and analyzed SNP polymorphisms (87–89) to characterize patterns of divergence and linkage disequilibrium across the region.

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## SUPPLEMENTARY MATERIALS

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Materials and Methods

Figs. S1 to S36

Tables S1 to S7

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MDAR Reproducibility Checklist

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